Shared decision making: Empathic encounters between design and healthcare

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Abstract This paper explores the role of design in the context of shared decision making (SDM). Through a reflection of the terms 'human centred design' and 'patient centred care', the negotiation of human beings' roles and experiences in both disciplines is highlighted. As a tool for reflection, this paper discusses a pilot study in SDM for breast cancer patients. The research shows that the systemic complexity of healthcare is well aligned to design's contemporary positioning as a social, participatory endeavour linking people as design partners. The project suggests that the meeting of the two disciplines will profit the most if mutual learnings in methods, language and approaches are seen as part of the added project value and are given room in the design process. Design brief, process and team are three key matters which require alignment towards a more rounded understanding of patients as co-creating the meaning of a healthcare system. By evidence of this pilot study, we argue that design has the means to objectify healthcare obstacles and might thus add tangibility to critical debates on a systemic level.

Keywords *Healthcare, Shared decision making, Co-design, Participatory innovation, People centred care*

Introduction

Traditionally, collaborations between healthcare and design focused on artefacts, see for example Bruce Archer's standard design for hospital beds (Archer 1964; Lawrence 2002), the attempt to *design out* medical error (West et al 2013) and design interventions that focus on the delivery of healthcare in the realm of co-designing services (Carr et al 2009; Freire et al 2010; Sangiorgi 2011). The complexity of manifold standpoints and the effort of negotiating new, participatory roles in healthcare such as required by SDM is an aspect that has had a surprisingly little amount of coverage in design research literature (Bowen et al 2013; Donetto et al 2015) and is consequently the focus of this paper. SDM aims at integrating patient's own preferences into treatment pathways, as a consequence individual need and mainstream systems meet.

Current decision-making models

SDM argues that healthcare professionals as the only parties with access to evidence is no longer plausible – instead SDM presupposes more equality, respecting patients values and healthcare professional's recommendation (see Légaré et al, 2014:6). Multidisciplinary research shows that involving patients in shared decision making can improve health outcome, patient satisfaction, adherence to treatment, reduces patient complaints and improves patients' knowledge and empowerment (see Légaré et al 2014).[1] This latest Cochrane Review also points to the pitfalls of the 39 studies reviewed: data on evaluation is scarce, consequently little is still known which activities work best. In SDM, approaches are diverse with two main variances: they either focus on the conversational situation between patient and clinician or offer an information system that can be used by patients themselves. Examples for the latter are option grids that guide patients through questions and lead to a profile with a suggested treatment option. Tools that focus stronger on the quality of the conversation between patient and clinician include Magic App (UK) (see Elwyn et al 2010) and cards and similar analogue decision aids developed by the Mayo Foundation for Medical Education and Research (US) (see Breslin et al 2008). From a designer's point of view, existing decision aids face difficulties on several levels: (1) Inclusivity, (2) a lack of a holistic integration of people's experiences and (3) their dependency on large-scale systemic dynamics. In terms of inclusivity, decision aids are typically heavily text based, as a digital source as well as on an analogue level. This strong reliance on reading abilities is critical given the diversity of people using healthcare systems. Secondly, it has been assumed that decision aids are best understood as helping the individual to find personally suited decisions. In fact existing aids target treatment options but individual experiences and socio-economic factors are hardly taken into consideration. It is known though that individual mobility or the level of family support, influences decision making. And third: Design and healthcare

1 Legare's et al's claims on shared decision making with patients, needs to be treated with some caution in terms of the contexts in which this is appropriate. It is for example important to highlight instances where this is inappropriate such as in psychosis.

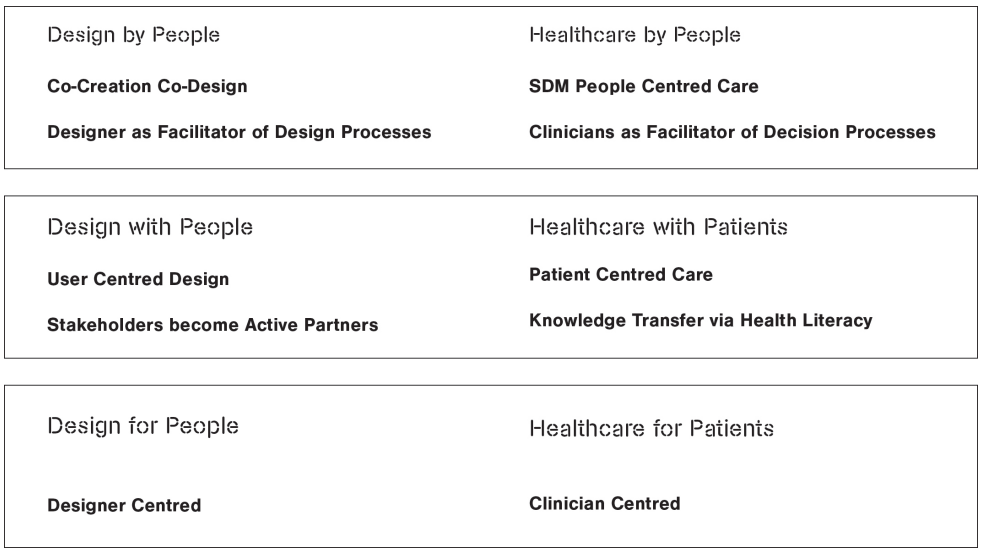


Figure 1. Through the lens of for, with and by.

distinguish different levels of participation (see next section), but in order to enable a transformative quality of participation, systemic dynamics cannot be solved on a local level. Communication skills of clinicians, the knowledge about SDM or the timeframe allowed for decision making mark a few of these systemic parameters.

Co-design as a re-definition of roles

Jane Fulton Suri (2007) offered the terms designing for, with and by people pointing to different relationships with people in the design process. The approach designing *for* perceives people as experts and consults them during the design process. Designing *with* people means a greater involvement of people in the design process, with stakeholders becoming increasingly active partners. Designing *by* people points to a shift of roles and responsibilities with designers becoming facilitators of design processes and people getting the authorization for making their own design decisions (Fulton Suri 2007). This most radical move in design practice calling for sharing tools and design knowledge with non-designers has also been reflected widely by contemporary design research (Simonsen&Robertson 2013; Sanders&Stappers 2008, 2014). Sanders and Stappers argue that whereas the design process is linear in user-centred approaches, roles gets mixed up in co-design: 'the person who will eventually be served through the design process is given the position of "expert of his/her experience" and plays a large role in knowledge development, idea generation and concept development' (Sanders&Stappers, 2008:12).

The equivalent shift in thinking in healthcare

When looking at healthcare through the lens of for, with and by people (Figure 1), we can see parallels of a strikingly similar development. Healthcare moved from a modernist idea of welfare and good healthcare provision for everybody to an attempt of moving tools and evidence to the patient and the general public. In the logic of this terminology, healthcare for patients focuses on problem solving and on the delivery of healthcare services being safe and effective; design projects in the area of patient safety and improving guidance systems in hospitals reflect this approach. Design is the supportive discipline in this respect, enabling goods, environments and services to be accessed by patients in the best possible way. Healthcare with patients can be linked to the idea of raising health literacy and transferring knowledge to the 'empowered and expert patient' (see Holmström&Röing 2010, Badcott 2005, Tyreman 2005, Civan and Pratt 2007 for a discussion on the limitations and advances of 'The Expert Patient').[2] Healthcare by patients would mark the most advanced development centering on the idea of shared decision making and patient democracy. While this participatory model is taken from design, parallel approaches in healthcare can be found in sources provided by the UK's Centre for Patient Leadership, the New Economics Foundation, NESTA, the King's Fund or the UK Design Council 'Double Diamond' Discover-Define-Develop-Deliver model for healthcare demonstration projects. The Centre for Patient Leadership for example offers individual patients as well as organisations, workshops with mentoring and coaching support, and focus on self-leadership, relationship building and dialogue, and developing influencing and advocacy skills, to enhance patient

2 In the history of healthcare the development of the terms 'person-centred' and 'client-centred' also relate to the work of the psychologist Dr. Carl Rogers (1902 – 1987). Initially developed as a form of psychotherapy, its principles of empathy and congruence were transferred to other areas such as patient care. The expert patient movement is one example for this development in its search for what constitutes a patient's expertise or to what extent a weaker notion of authority would support a more egalitarian system. The limitations and advances of a non-judgmental and empathic relationship in healthcare correlate with SDM.

engagement.[3] The skills sharing approach can be compared to passing on tools in co-creation or co-design processes.

SDM through the lens of participatory innovation

The focus of healthcare and design shows that both disciplines share a similar development from doctor and designer centred to a people centred approach, from closed specialist processes to open ones. However, cross-disciplinary collaboration bears both real-world and conceptual questions. Participatory design/innovation offers a framework that considers the diversity of language, conversation styles and methods as windows of opportunities and is thus discussed as a potential lens for approaching collaborations between design and clinical healthcare.

'Boundary' difficulties, opportunities and empathic encounters

Studies in design and innovation management reflect on various challenges met by corporations and non-profit organizations trying to implement more open and participatory approaches to innovation. Carlile (2002) argued that conflict arises at knowledge boundaries, meaning that knowledge is highly localized and embedded, causing conflict when working across functions. Stacey and Griffin (2005) see organizations in a steady interplay between inclusion and exclusion of different stakeholders with different identities and beliefs. In line with this discourse in organizational management, Sproedt and Larsen claim that '(…) the innovation process emerges not as a result of one singular idea or intention, but in the meeting of differences' (Sproedt&Larsen, 2012:1004). Fonseca (2002) explains in this context that innovation is an ongoing process of understanding and misunderstanding project partners. The emergence of conflict is thereby perceived as opportunity, spotlighting constraints and 'conversations in the shadow' (see Larsen&Bogers 2014) as a productive source of novelty and innovation, rather than as failed approaches of co-operations. 'In the local, emerging interaction people negotiate meaning of knowledge and make sense of novelty' (Sproedt&Larsen, 2012:1005). Via the pilot study covered in this paper we reflect on some of the themes discussed in participatory innovation and do so by observations on a collaboration between a Danish hospital group and the design development department 'Social Inclusion'[4] of the first author's design institution.

Table 1. Methods.

Project Phase	Methods	Time Frame
Collect	Project-Setup, Participant Observation, Patient Interviews, Desktop Research	2,5 Months
Comprehend	Analysis, Visualizing Data, Personas	1 Month
Conceptualize	Co-Creation Workshop, Prototyping	1 Month
Create	Communication Design, Testing	0,5 Month

Insights into how design and healthcare could collaborate in the best possible way are searched for along these research questions: What are the challenges and opportunities at the boundaries between different disciplines learning to work together (here design and clinical healthcare)? What role does participatory design play in the shift of power from top-down healthcare systems to one more democratically involving those individuals affected by decisions made? How could design support SDM in integrating patients' (often) emotional experiences into treatment pathways?

Method

This paper is based on a pilot study in SDM in breast cancer. The methods and process outline base on the human centred '5C MODEL' (Fries&Gelting 2014). The phases 'Collect' and 'Comprehend' follow the contemporary understanding of a human-centered design process including an observational phase to meet stakeholders in the context of the specific project (see also Blomberg et al 1993; Blomberg&Burrell 2009). 'Conceptualize' and 'Create' point to ideation and innovation phase (the fifth C 'Collaborate' forms a connecting theme). (Table 1)

The design work has been carried out by the design development team, while the first author of this paper had an observer role. The design team observed 36 patient-doctor consultations at the hospital. Participant observation was followed by 15 semi-structured interviews focusing on treatment plans, information and options. The design team's desktop research focused on brochures and information material that patients usually receive from the hospital. (Table 2)

The limitations of this pilot study have to be acknowledged fully on several levels: due to the hospital's clinical pathways only eight patients of the observed 36 formed part of the target patient group (newly diagnosed breast cancer patients offered post-operative prophylactic chemotherapy and/or anti-hormone therapy to help prevent disease relapse). A questionnaire was sent out to involved physicians, nurses and the design team by the design researcher after the end of the five-month project, informing them that responses will be anonymized and used to gather project learnings for shaping potential future

3 Doughty, one of the founders of The Centre for Patient Leadership says: "It's about building effective relationships, being able to communicate and having the ability to dialogue ... and talking together about problems that you share. We are about giving patients the confidence and the skills to engage with other professionals in which the patient themselves perceive themselves as being equal." (The Guardian, accessed 20 March 2016).

4 The design development departments at Design School Kolding focus on three labs, the Lab for Social Inclusion focuses on prevention and welfare technology (see https://www.designskolenkolding.dk/en/lab-social-inklusion).

collaborations. The questionnaire used open format questions to give the respondents the opportunity to express their opinions in an unrestricted mode, including open-ended questions seeking suggestions for improvements in future projects.

While all three designers responded, only one clinician gave formal feedback. Patient satisfaction has not been evaluated because the tool has not been implemented yet.

The paper aims for analytical generalization by reflecting on notes and experiences in search for patterns of interaction between the collaborative partners. The following sections describe the design process in chronological order as observed by the authors of this paper; they therefore represent subjective viewpoints.

A tool for shared decision making in breast cancer

Healthcare demographics in Denmark

The collaborative hospital partner is a cancer and university centre in Denmark planning to gain expertise and knowledge on SDM and to establish a knowledge centre within the hospital group. In the past few years, hospitals in Danish healthcare have seen increasing centralization and specialization (Olejaz et al 2012). 'In this attempt to optimize and standardize treatment and care in large specialized hospital units, the individual patients' needs and preferences may be lost' (Dahl Steffensen, 2015:3). A baseline study commissioned by the hospital underlines that cancer patients want to participate in decisions about their treatment. According to the survey every fifth lacked the opportunity to participate and healthcare professionals pointed to the absence of tools to perform SDM during their consultations (Olsen et al, 2013 a and b). In respect to cancer patients who generally receive potent and potentially hazardous treatments involving various healthcare settings and several healthcare professionals '(…) this makes it essential to discuss with the individual patient potential harms and benefits of different treatment options, including the option not to treat' (Dahl Steffensen, 2015:3).

Table 2. Stakeholder Chart.

Stakeholders	Number and further Specification
Patients	36 (with 8 from main target group)
Clinicians	10 (5 oncologists, 5 nurses)
Designers	3 (with specialization in communication, service and product design)
Design Researcher	1 (specialized in welfare design)
Steering Group	7 (Head of Innovation, Research Management, Healthcare Researchers, Clinicians, Director of Centre of Shared Decision Making)
Administrative Worker	1 (admin background)

The sesign process: From kick-off to final presentation

Project pre-phase

An initial planning group suggested working on a tool for guiding patients' decisions on whether to participate in a clinical trial or to receive standard treatment. Administration changed this initial focus to the current task, described below. Additionally to a change of project focus, the collaboration started before the knowledge center for SDM was established and obtained knowledge.

Project kick-off

The design brief for the first collaboration in the field of SDM between hospital and design school focused on the development of a decision tool for use at the Department of Oncology in their work with newly diagnosed breast cancer patients. The project's time frame was five months. In the kick-off meeting, project leaders from Design Development Department and the hospital, the innovation manager and clinicians were present. The meeting centred on framing the working process based on the briefing received from the hospital. The project leader from the design school focused on the process such as access to patients and clinicians as a basis for a user-centered design process. The director of the SDM knowledge center emphasized her wish for a tool implementable in clinical pathways: 'I don't want this tool to sit on the shelf. I want it to be used in the clinic.' There has been no discussion at this point about sufficient time to meet this goal and no issues around the content of the design brief were raised. The design project leader has developed a design process description based on the meeting.

Month 2 Skype interview with one of the designers

The status quo of the project was discussed in a talk between the first author of this paper and one of the designers. From the designers' point of view the project steps and responsibilities seemed straight forward as planned in the design brief. Nevertheless uncertainty arose when concrete appointments were to be made for interviews with stakeholders, more precisely with cancer patients. The designer reflected on these challenges with the planned interviews:

Nobody was in charge of telling the patients. (Designer#1)

While contacts to clinicians had been established at the first meeting and beyond, the contact to patients was out of reach for the design team, they relied on the hospital to organize them. In the same talk, the designer reflected on first findings and design opportunities, thus at this point of the project the design team had participated in 15 patient consultations.

They [the patients] want the doctors to decide. (…) Nobody says no to treatment. There is too much information, so what is missing is a tool to remember information. (Designer#1)

Based on these observations, the mismatch of design



Figure 2. Ideation of three dimensional 'changeable roadmap'.

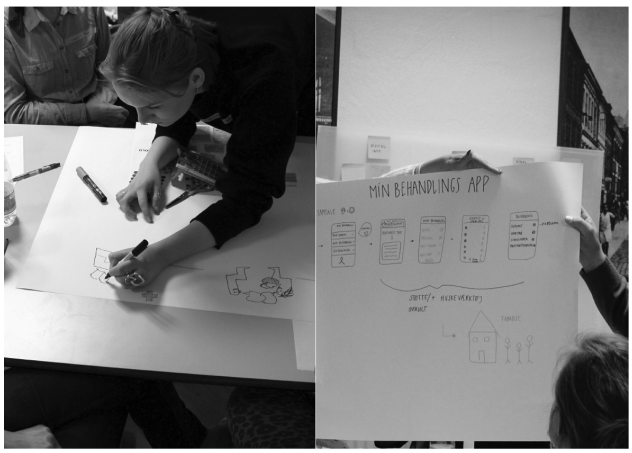


Figure 3. Generating ideas at the workshop.

brief and field observations became obvious for the design team. These consultations aimed at post-operative prophylactic therapies given to lower the risk of the cancer coming back. These therapies preventing a cancer relapse offered only two options – treat or not treat. Following the logic of SDM and the design brief, 'treat' or 'not treat' would have been the two options for the decision tool. One of the main design opportunities was though located as a tool capable of documenting the information given at the consultation.

Month 3 Presentation of three concepts
The focus of this meeting was the presentation of the design process thus far and a discussion on how and with which concept to proceed. This time, participants also included the steering group. Via an analysis of designer's observations, twelve themes ranging from 'horror stories', 'information' or 'letter from the hospital', 'a lot of words', 'preparation', 'common agenda', 'which info when' were clustered. During the presentation the design team supported the themes with quotes from their interviewees. In order to make the types of patients more tangible, the design team worked with three personas: 'Vera, 61, positive', 'Mette, 44, confused' and 'Henriette, 36, critical' were chosen as representatives for typical patients and their experiences, emotions and socio-economic background, their trust in the clinician, how they react to the consultation and if they brought relatives with them to the meeting or not. The method of personas, that considers social-economic aspects of the design target group (a common but also criticized tool in design practice, see Tonkinwise 2011), was recognized by the clinicians as a precise rendering of their patient group. Finally the design team presented the spaces of opportunity organized around the three concepts: preparation tool, common agenda and remembering tool. The debate among meeting participants was on whether to focus on a qualification tool ('preparation') or a remembering tool. One idea vividly discussed amongst the meeting members focused on a 'changeable roadmap' (Figure 2), a tool that could start from an empty road at the beginning of the consultation and be filled up with treatments and procedures as agreed upon between clinician and patient. The clinicians as well as other meeting members added ideas to this concept.

The design team tested the idea of the 'changeable roadmap" with three-dimensional bricks but regarded a game-like tool as inappropriate for the subject of cancer in an adult patient group.

Month 4 Co-creation workshop
The workshop took place at the hospital and brought together patients and relatives, three doctors and three nurses; it was chaired by the design team and supported by design students that helped facilitating and visualizing the process and group work (Figure 3).

After presenting the schedule, the designers shared their process and findings with the workshop team (in a similar way as the presentation at the steering group meeting), afterwards mixed teams and discussed questions such as: 'How do you define shared decision making?' and 'Please give an everyday example of SDM.' As the next task team members were asked to individually note down their thoughts on the three basic design concepts so far, a preparation tool, common agenda and remembering tool. After a presentation in symposium, every member could choose a favourite concept to proceed with for an individual idea brainstorm focusing on wishes and ideas for the specific concept. All ideas were clustered and structured into one big poster. The team members were asked to vote again for one of the generated ideas; the 'winner' was discussed in more depth, focusing on form and content of a potential solution. Finally one design student each drew and visualized their team's solution showing how the idea could work step by step. A presentation and discussion followed. Group One developed the idea of a 'Patient Bag' that provides the road map for the patient journey ahead. Group Two proposed an app that visualized the treatment pathway and Group Three presented symbolic representation of the therapies divided in three categories. What united all three approaches was the support for cancer patients with information that might eliminate some of the emotional concerns, questions and uncertainties.

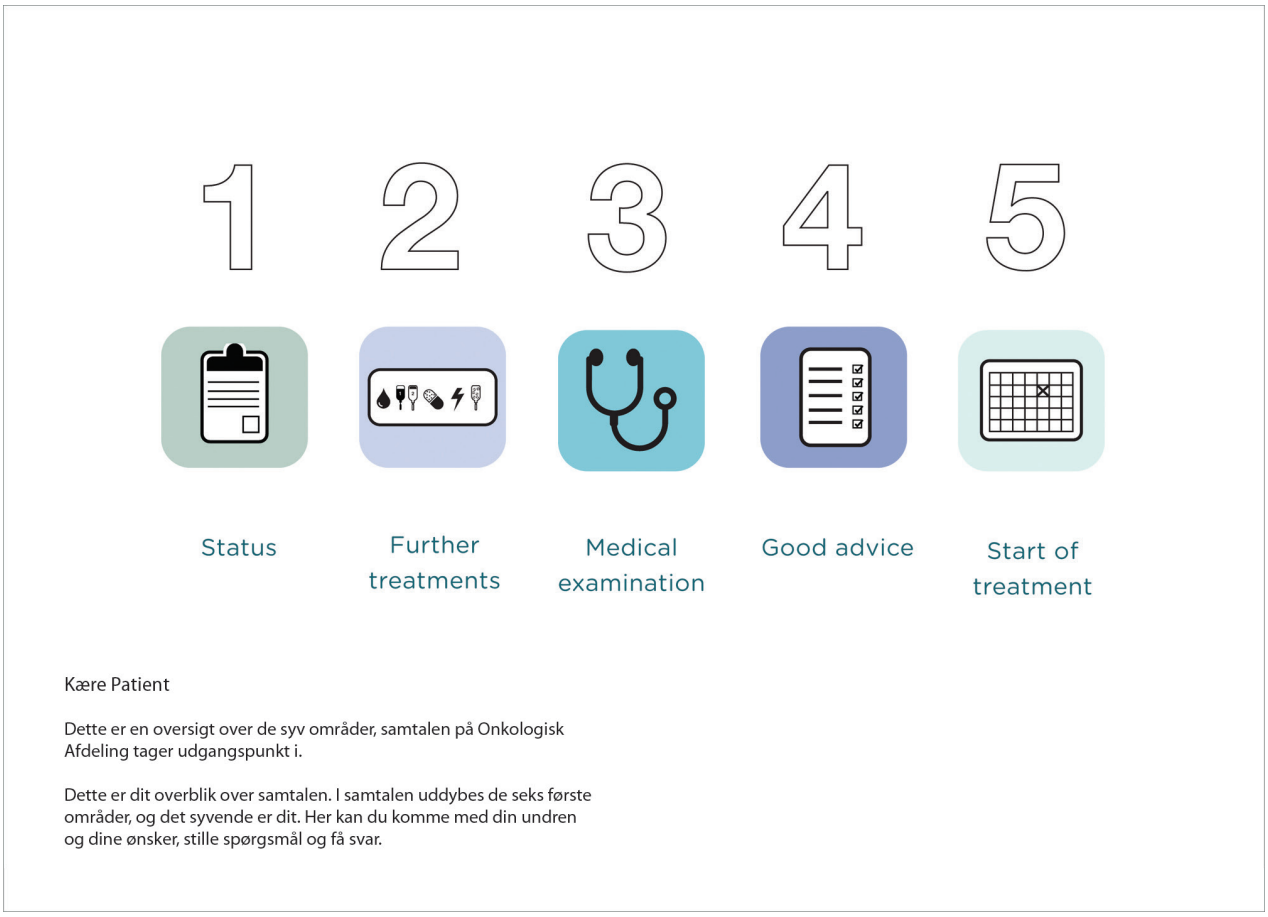


Figure 4. The introduction sheet.

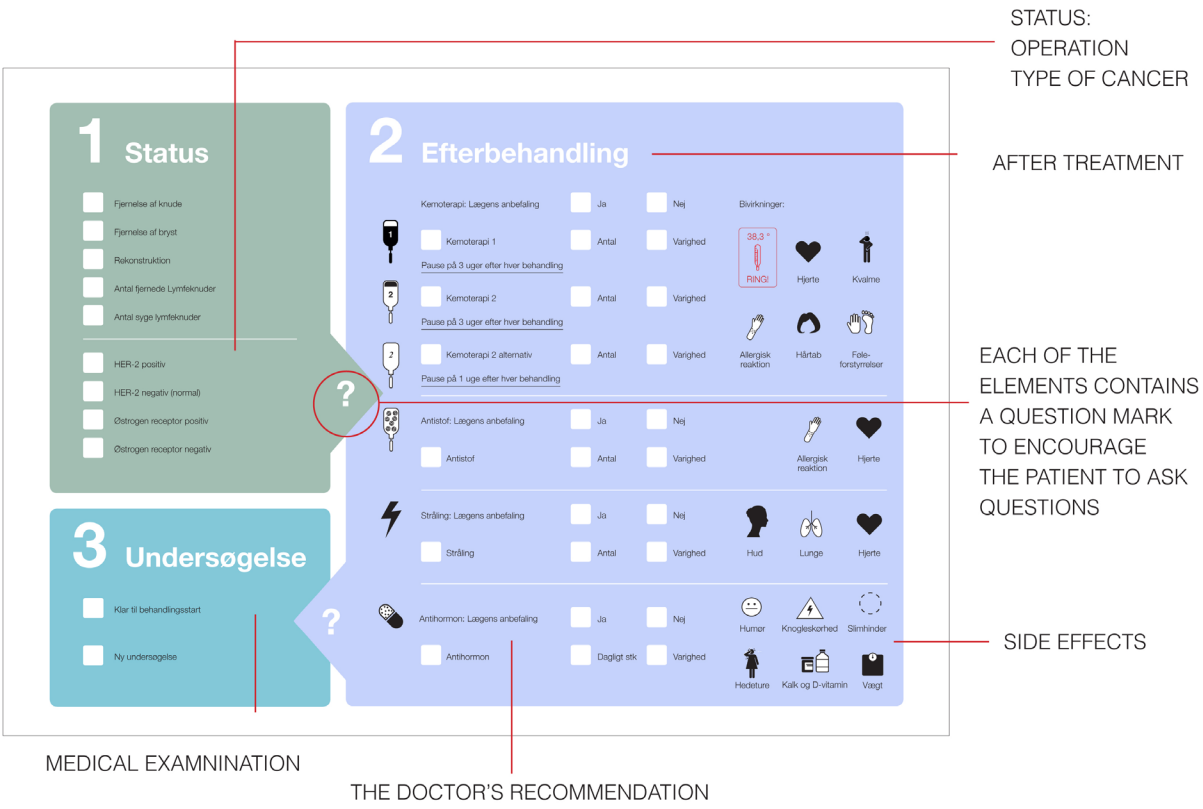


Figure 5. Conversational sheet 2_oncologist's page.

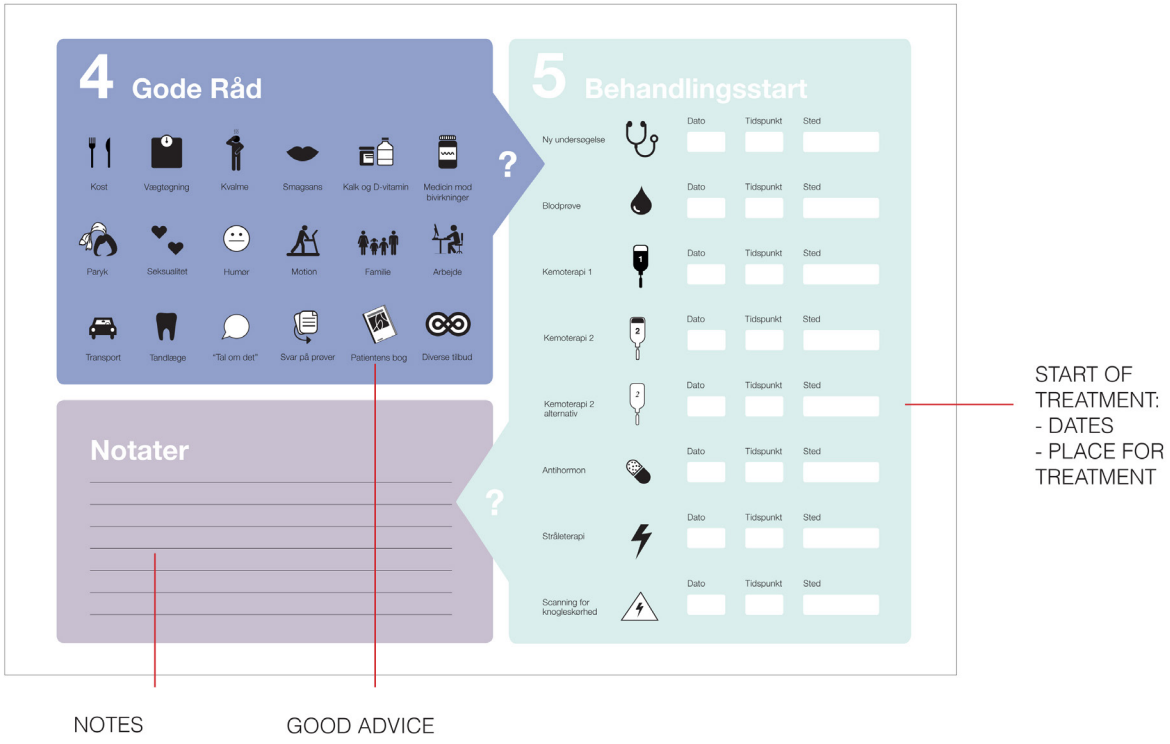


Figure 6. Conversational sheet 2_nurse's page.

Month 5 Final presentation
Based on the analysis of fieldwork and co-creation workshop the team decided to prototype a tool that shall unite a preparation, qualification, conversation and checking tool in one. To be used during the physician consultations this first proposal consists of an A4 introduction and two A3 conversation sheets. Before the consultation, the patient receives the introduction sheet which provides a visual overview of the elements of the consultation (Figure 4) along with the notice letter. These five main areas cover health status (1), further treatment (2), medical examination (3), good advice (4) and start of treatment (5). During the consultation, clinicians and patient cover the five consultation elements using the two conversation sheets (Figure 5 and 6). Doctors, nurses and patients shall fill out the sheets together and thereby actively share information. The final proposal is also a means for communicating and informing family and kin in a brief way – a decisive aspect in a socially challenging life phase. The final proposal is also a means for communicating and informing family and kin in a brief way – a decisive aspect in a socially challenging life phase.

Analysis

The data that has been acquired rests on knowledge from the design process, observations by the design researcher following the process, as well as the analysis of the questionnaire sent out to designers and clinicians after the project. The limitations of this pilot study have been discussed in the method's section, the following analysis should therefore be viewed as learnings for a second study following this one. The three main areas of interest for emerging design opportunities are explored in the following:

The design brief and its relation to the outcome
When analysing the project and looking at the brief, it becomes evident that it frames the task very precisely but that there is a mismatch between brief and actual clinical encounter, they did not harmonize. The questionnaire quotes from both clinicians and designers support these observations:

(…) the aim of this project was changed centrally from the administration (…) the decision making situation was changed to a situation where the patient do not have a real choice between several options – it was instead a treat or not to treat decision. (Clinician)

The goal of the project did not fit into the real world, meaning that shared decision takes place in another part of the patients journey. (Designer#2)

The fact that a design brief does not represent the most demanding problem is a well-known aspect in many design projects. User centred design thus bases its concepts on fieldwork, which this project also did. This approach indicates that design briefs frame the desired direction of the outcome but they do not prescribe the medium or form because this can only be decided in the course of the project. The more precise picture the design team gains, the clearer the problem becomes, and the more clearly a possible solution can be described. Design research discusses this as the 'co-evolution' of problem and solution (see Rittel&Webber 1973 for an early discussion of 'wicked problems' and Buchanan 1992 and Coyne 2005 for more recent ones). A crucial finding is that the tool targets primarily the recall of medical information rather than the decision making process. The medical decision has to be been taken earlier in the patient

journey. The current first proposal is an objectification of the patient journey in this given health care system. An enhanced permeability from bottom-up (instead of a top-down understanding of health care systems) is thus eligible. Small-scale observations as made in this pilot study, might point to vital systemic obstacles (e.g. the time allowed for decision making). Participatory design loose strength, if broader systemic topics are not questioned more widely.

The design process

Designers all pointed to the tight time frame that did not foresee enough resources for testing; the clinician also recognized a lack of touchpoints but did not link it with a problem in the set timeframe but rather the team set-up. This lack of time is also reflected on at several points in the questionnaires:

I am a little worried about the next phase, no time or considerations have been made about the implementation process. (…) Next time there should be a strategy for execution of the designed product.

Producing prototypes is the best tool for understanding the end product. These processes take time, and we needed more time to test. (Designer#2)

After the stakeholder workshop which took place after the Concept Presentation, no further meeting was scheduled and no time was reserved for an implementation phase with testing and iterative design adaptations. The meeting structure between project steering group and design team took place in the following way (Figure 7):

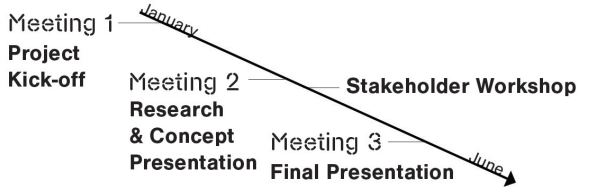


Figure 7. The meeting structure.

By evidence of this pilot study, future collaborations' time frame should be more realistic. The 5C Model used by the design team lacks an implementation and evaluation phase. As part of a more robust study design for the following project, this model shall thus be complemented by participatory process models (Simonsen and Hertzum 2012) more effective in health care settings.

The team and missing resources

The analysis of the process points to further human resources' positions that would have been beneficial. For sustainable future projects, more resources should be ensured for integrating learnings from existing international projects as well as supporting evaluation, implementation and dissemination with a research perspective. This research resource would better frame the concept of SDM and equip the design team with an advanced knowledge base. Although there was a third party administrative worker assigned, this project showed that it needs a clinical insider to coordinate stakeholders and, more importantly, to communicate the project vision internally and externally.

When asking for missing competencies in the project set-up in the questionnaire, the clinician puts it forward in the following way:

Implementation team and team for sustaining the project (…). Ideas for future perspectives – what are the next steps? How to measure and evaluate whether the product has made a difference to the patients and to the clinicians. (Clinician)

A health researcher providing a system for evaluation and measurement should be included in future projects to ensure an active learning culture alongside the design researcher.

Everybody in the design team mentioned the emotional weight of being confronted with cancer patients' individual cases. Providing the opportunity for mediation sessions seems an evident necessity to allow for reflection and supervision of designers working with cancer patients.

Discussion

What can this pilot study demonstrate for collaborative encounters between design and SDM in healthcare on a wider scale? A collaborative project set-up can be viewed as a tool for knowledge transfer and this is inherently ridden with constraints, including the messiness of multiple viewpoints. Experience-based design in SDM should therefore acknowledge disagreement and conflict as productive part of a new 'anthropology of services" (see Blomberg and Darrah 2015). While SDM can be an object of design, it likewise requires the acknowledgement of design's boundaries. With SDM's nature of requiring enactment by all stakeholders, it expands the character of a finite service or product. This asks for an open-ended live lab and partial results rather than finished objects. In this respect a clear focus on implementation as an iterative process is essential and shall be ensured by robust study designs.

The greatest social gains lie in an ongoing exploration of the concept of shared decision making together with stakeholders. Mattelmäki et al (2011) argue for the significance of 'incompleteness of field study outcomes as it invites sense-making through (…) empathic understanding and engagement'. In a similar sense collaborative design is 'a process of exploring what it is that will create value for specific people' (Mattelmäki et al, 2011:79). New design briefs should include a collective exploration of the questions: What is the meaning of SDM in the local context? Healthcare research has the means to frame this definition and understanding; communication design can convert this into a strong identity supporting a collective ownership through a joint stakeholder narrative and experience. By evidence of this pilot study we regard this strategy as essential, thus local healthcare systems and people within the institutions determine if concept are used and implemented sustainably.

The most essential discussion and learning in this pilot study regards the interplay of local and large-scale dynamics in SDM. The project objectified one specific aspect of the current patient pathway and made it tangible through a first design proposal: Although the design brief asked for a decision tool, fieldwork showed that the medical decision is taken at another point on the clinical pathway. This fact raises questions such as: Is the time and support appropriate that is currently allowed for patients to make far fledged decisions? Are healthcare systems balancing the quest for mainstream pathways well with the individuality and experiences of its patients? What this pilot study shows is that the obstacles and misunderstandings of the two disciplines working together gave a clearer picture of large-scale systemic problems rather than only those on a local level. Patient activation, communication skills of clinicians, the level of knowledge about SDM in the wider public, mainstream healthcare pathways – they all challenge the application of SDM. Consequently, findings point to a need for tools that allow for systemic modification and change in healthcare. The difficulties related to this pilot study are thus symptoms, in need for careful examination towards a more experience driven practice. Future study designs need to be aware of notions of co-realization and discourses of infrastructuring in participatory design (see Star & Ruhleder 1996; Hartswood et al 2008, Karasti 2014).

Conclusion

Shared decision making is redefining a historically learned role-setting in hospitals. These shifts in status and power between clinicians and patients request a fundamental change in our understanding of healthcare in general. It calls for a swing of mind-sets in healthcare – ideally performed by all stakeholders connected to it. Participatory, empathic design acknowledges misunderstanding and conflict as the basis for novelty and advancement. However the power of participatory concepts such as SDM will be measured by its strength to impact healthcare systems' set structures more fundamentally from bottom-up, based on people's actual experiences.

This paper discussed the development of human centred design as a potential lens for looking at patient centred care. In design, *users* turned into *people* and *citizens*, based on a more holistic idea of working with people as co-creators of services and products. If healthcare aspires to implement shared decision making as a future standard, a shift in terminology towards *people* centred care would follow consequently. We call for further cross-disciplinary research into the practical, conceptual and political implications of the role of the patient in SDM. What methods support the involvement of emotional experiences in SDM is suggested as a potential area for future research in Design&Emotion. Critical evaluation of the collaboration between design and healthcare should consequently focus on the swing and constitution of power relations as esteemed in SDM.

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